Krishna Koushik Maddukuri

Senior Software Engineer | Java · Spring Boot · Apache Kafka · AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience designing, building, and operating large-scale distributed systems and cloud-native microservices for financial services firms including Bank of America and Capital One. Expert in Java, Spring Boot, Apache Kafka, REST API design, system design, and Amazon Web Services (AWS), with a record of delivering secure, regulatory-compliant, high-throughput data platforms. Reduced batch processing time by 43%, cut deployment time by 70%, and lowered compute costs through container optimization. AWS Certified Developer – Associate with deep experience in event-driven architecture, CI/CD automation, test-driven development (TDD), and production observability.

TECHNICAL SKILLS

Languages: Java (8/11/17), Python, SQL, TypeScript, JavaScript, C++, Bash, Groovy

Backend & Frameworks: Spring Boot, Spring MVC, Spring Batch, Spring Kafka, Spring Security, Spring Data JPA, Hibernate, Microservices, Distributed Systems, System Design, RESTful APIs, OpenAPI/Swagger, GraphQL, OAuth2, JWT, Node.js, Liquibase

Cloud & Infrastructure: AWS (ECS, Fargate, EC2, Lambda, S3, SNS, SQS, RDS, DynamoDB, Step Functions, CloudFormation, CloudWatch, IAM, Glue, EMR), Microsoft Azure, Docker, Kubernetes, Serverless Architecture, Infrastructure as Code (IaC)

Data & Messaging: Apache Kafka, PostgreSQL, MySQL, SQL Server, DynamoDB, MongoDB, Event-Driven Architecture, Real-Time Data Pipelines, ETL, Metro 2 Credit Reporting

DevOps & CI/CD: Jenkins, Git, Bitbucket, GitHub, Maven, Gradle, SonarQube, Blue-Green Deployments, Zero-Downtime Releases, Agile/Scrum, Jira, Confluence

Testing & Quality: JUnit, Mockito, Cucumber, Selenium, WireMock, Postman, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Code Review

Observability & Security: Splunk, New Relic APM, Prometheus, ELK Stack, Spring Actuator, Log4j, Performance Tuning, Secure SDLC, Regulatory Compliance

Frontend: React.js, Redux, Angular, HTML5, CSS3, Bootstrap, jQuery

PROFESSIONAL EXPERIENCE

Application Programmer III (Software Engineer) **Oct 2026 – Present**
Bank of America *Pennington, NJ*

- Develop and maintain Java/Spring Boot microservices and RESTful APIs for Bank of America's Wealth Management Technology platform, delivering secure, scalable features for high-availability financial applications.
- Build event-driven integrations with Apache Kafka and relational data stores, following the bank's secure SDLC, code review, and regulatory compliance standards.
- Contribute across the full stack with automated unit and integration tests (JUnit, Mockito) and CI/CD pipelines, collaborating in Agile/Scrum ceremonies with product, QA, and platform teams.

Senior Software Engineer **Sep 2024 – Oct 2026**
Capital One *McLean, VA*

- Designed and built scalable, secure Java/Spring Boot microservices for Capital One's Enterprise Credit Bureau Reporting Service, generating compliant Metro 2 files for approximately 30 million consumer, small-business, and commercial accounts each month across Equifax, Experian, TransUnion, and LexisNexis.
- Engineered high-throughput, event-driven pipelines with Apache Kafka, Spring Kafka, and PostgreSQL on Amazon RDS, achieving fault-tolerant, exactly-once processing of sensitive financial data at enterprise scale.
- Improved batch performance by 43% (7 hours to 4) through performance tuning of Spring Batch readers, replacing cursor-based readers with a paginated reader for high-volume Metro 2 file generation.
- Deployed containerized services on AWS Fargate via CloudFormation, integrating SNS, S3, RDS, and Step Functions; reduced container image size by 60%, lowering compute costs and improving runtime efficiency.

- Implemented enterprise CI/CD pipelines in Jenkins to automate builds, enforce SonarQube quality gates, and run blue-green deployments with zero-downtime releases.
- Strengthened security and compliance by designing OAuth2-protected REST APIs and building automated test suites with JUnit and Cucumber, reducing production defects.
- Integrated Splunk and New Relic APM with actionable dashboards and alerting, accelerating incident response and root-cause analysis across distributed microservices.
- Onboarded Discover account portfolios into the credit bureau reporting pipeline following the Capital One–Discover merger, extending the platform to new product segments without disrupting monthly regulatory deadlines.

Software Engineer **Sep 2023 – Sep 2024**
NGA Group Inc *New Brunswick, NJ*

- Designed a cloud-ready Java/Spring Boot service unifying disparate data sources into a standardized reporting model, improving data accessibility and eliminating manual reconciliation across analytics workflows.
- Delivered secure, role-based REST APIs enabling downstream teams to consume curated datasets without direct database access, strengthening data governance.
- Built a high-velocity ingestion pipeline with Apache Kafka and DynamoDB, supporting near-real-time propagation of updates and scaling with increased upstream event volume.
- Containerized the application on AWS ECS with Docker and automated build and release pipelines in Jenkins, reducing deployment time by 70% and improving release reliability.
- Improved operational visibility by integrating Spring Actuator, Prometheus, and ELK analytics, cutting issue-detection time by 50%.

Software Engineer **Aug 2020 – Aug 2021**
Infosys Limited *Hyderabad, India*

- Developed modular, reusable React.js components with Redux state management, delivering consistent UI behavior across 4 large-scale enterprise applications for a global retail client.
- Integrated RESTful APIs with React and Redux for asynchronous data fetching and built front-end features (forms, routing, events, custom components) backed by Java/Spring MVC services.
- Modernized legacy codebases and applied engineering best practices, reducing bug rates by 25% and improving maintainability and performance.
- Implemented CI/CD pipelines with Bitbucket and automated multi-region AWS deployments, increasing deployment frequency by 3x.

Software Engineer Intern **Jan 2020 – Aug 2020**
Infosys Limited *Mysore, India*

- Completed 15+ projects and certification-aligned exercises across back-end and front-end development within the MERN ecosystem (MongoDB, Express, React, Node.js).
- Diagnosed and resolved backend build failures by analyzing Jenkins logs and debugging deployment scripts in Bash and Groovy, improving CI/CD pipeline stability by 40%.
- Coordinated sprint planning for a four-developer team using Agile methodology and Jira, ensuring on-time delivery of project milestones.

EDUCATION

Doctor of Business Administration (DBA), Technology Management Jul 2026 – Aug 2029 (expected)
Westcliff University

Master of Science (M.S.), Computer Science 2021 – 2023
George Mason University, Fairfax, VA

Bachelor of Technology (B.Tech), Computer Science 2016 – 2020
Amrita Vishwa Vidyapeetham, Coimbatore, India

CERTIFICATIONS

- AWS Certified Developer – Associate, Amazon Web Services (issued Nov 2023)
- AWS Certified Cloud Practitioner, Amazon Web Services (issued Jun 2023)